

Time Series Exercise Sheet 7

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Exercise 7.1

Let $\{\varepsilon_t\}_{t \in \mathbb{Z}}$ be a white noise process with variance σ_ε^2 . Determine the spectral density function.

Solution 7.1

If ε_t is a white noise, then the process is uncorrelated so $\gamma_\tau = \sigma_\varepsilon^2$ if $\tau = 0$, and 0 otherwise. Therefore, the spectral density of $\{\varepsilon_t\}$ is for all f

$$S(f) = \sigma_\varepsilon^2.$$

Exercise 7.2

Consider the sequence

$$\gamma_\tau = \begin{cases} 1 & \text{if } |\tau| \leq K \\ 0 & \text{if } |\tau| > K \end{cases}$$

where K is a positive integer. Is γ_τ an ACVS for some discrete parameter stationary process $\{Y_t\}$? What would be the SDF?

Solution 7.2

Note first that $\sum_{\tau \in \mathbb{Z}} |\gamma_\tau| < \infty$ and $\gamma_\tau = \gamma_{-\tau}$, and that we can define the function $S(\cdot)$ such that for all $f \in [-1/2, 1/2]$

$$\begin{aligned} S(f) &= \sum_{\tau=-\infty}^{\infty} \gamma_\tau \exp(-2\pi i \tau f) \\ &= \sum_{\tau=-K}^K \exp(-2\pi i \tau f) \\ &= 1 + 2 \sum_{\tau=1}^K \cos(2\pi \tau f) \end{aligned}$$

However, this does not satisfy $S(f) \geq 0$ for all $f \in [-1/2, 1/2]$, so γ_τ cannot be an ACVS for a stationary time series. Indeed, if K is odd, i.e. $K = 2q + 1$ for $q \in \mathbb{Z}$, we have $S(1/2) < 0$.

Exercise 7.3

Determine the auto-covariance function of $\{X_t\}_{t \in \mathbb{Z}}$ if it has spectrum

$$S(f) = \sigma^2 \frac{(1 - 2|f|)}{\pi} \mathbb{1}(|f| \leq 1/2)$$

Solution 7.3

We compute for $|\tau| > 0$:

$$\begin{aligned}
 \gamma_\tau &= \int_{-1/2}^{1/2} e^{2\pi i f \tau} S(f) df \\
 &= \frac{\sigma^2}{\pi} \int_{-1/2}^{1/2} e^{2\pi i f \tau} (1 - 2|f|) df \\
 &= \frac{\sigma^2}{\pi} \int_{-1/2}^{1/2} (\cos(2\pi f \tau) + i \sin(2\pi f \tau))(1 - 2|f|) df \\
 &= \frac{\sigma^2}{\pi} \int_{-1/2}^{1/2} \cos(2\pi f \tau) (1 - 2|f|) df && (\sin \text{ is odd}) \\
 &= \frac{2\sigma^2}{\pi} \int_0^{1/2} \cos(2\pi f \tau) (1 - 2f) df && (\text{Integrand is even}) \\
 &= \frac{2\sigma^2}{\pi} \left\{ \underbrace{\int_0^{1/2} \cos(2\pi f \tau) df}_{I_1} - 2 \underbrace{\int_0^{1/2} \cos(2\pi f \tau) f df}_{I_2} \right\}.
 \end{aligned}$$

Then $I_1 = \frac{\sin(\pi \tau)}{2\pi \tau}$, and after computing the second integral $I_2 = \frac{\sin(\pi \tau)}{4\pi \tau} + \frac{\cos(\pi \tau) - 1}{4\pi^2 \tau^2}$. Summing up both we end up with

$$\gamma_\tau = \frac{\sigma^2}{\pi^3} \left\{ \frac{1 - \cos(\pi \tau)}{\tau^2} \right\}.$$

For $\tau = 0$, we compute $\gamma_0 = \frac{\sigma^2}{2\pi}$.

Exercise 7.4

What is the spectral density function of an MA(q) process?

Solution 7.4

Recall that for an MA(q) we have autocovariance function given by

$$\gamma_\tau = \begin{cases} \sigma_\varepsilon^2 \sum_{j=0}^q \theta_j \theta_{j+|\tau|} & \text{if } |\tau| \leq q, \\ 0 & \text{otherwise,} \end{cases}$$

where $\theta_j = 0$ for $j > q$. In fact, looking at how this was computed we have the more useful equation:

$$\gamma_\tau = \sigma_\varepsilon^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \mathbb{1}_{\tau=k-j}.$$

Clearly $\gamma \in \ell^1$, so the spectral density function exists and can be computed by taking the Fourier transform of the ACVS. Thus the spectral density function is

$$\begin{aligned}
 S(f) &= \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i \tau f} \\
 &= \sum_{\tau=-\infty}^{\infty} \sigma_\varepsilon^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \mathbb{1}_{\tau=k-j} e^{-2\pi i \tau f} \\
 &= \sigma_\varepsilon^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k e^{-2\pi i (k-j) f} \\
 &= \sigma_\varepsilon^2 \left| \sum_{j=0}^q \theta_j e^{-2\pi i j f} \right|^2.
 \end{aligned}$$

Exercise 7.5

Consider a harmonic process with a time varying amplitude

$$X_t = \epsilon_t \cos(\nu t + \Theta)$$

where we assume that ν is fixed, Θ is a random phase, and ϵ_t is not constrained to be positive, but that its a mean-zero white noise process with variance σ^2 . Moreover, assume that ϵ_t is independent of Θ . The harmonic process in the lecture stipulated that Θ had to be uniformly distributed on $[-\pi, \pi]$. Suppose (instead) that Θ has a pdf given by

$$f_{\Theta}(\theta) = \frac{1}{2\pi} (1 + \cos(\theta)), \quad |\theta| \leq \pi$$

Determine the first two moments of X_t . Are they still independent of t ?

Solution 7.5

Using the independence between ϵ_t and Θ , we have

$$\mathbb{E}[X_t] = \mathbb{E}[\epsilon_t] \mathbb{E}[\cos(\nu t + \Theta)] = 0 \cdot \mathbb{E}[\cos(\nu t + \Theta)] = 0$$

which is constant in t , as the second expectation is finite since $\cos$ is bounded above by unity. Again we have for all $t, \tau \in T$

$$\mathbb{E}[X_t X_{t+\tau}] = \mathbb{E}[\epsilon_t \epsilon_{t+\tau}] \mathbb{E}[\cos(\nu t + \Theta) \cos(\nu(t + \tau) + \Theta)]$$

Since ϵ_t is uncorrelated, we have $\mathbb{E}[\epsilon_t \epsilon_{t+\tau}] = \sigma^2$ if $\tau = 0$, and 0 if $\tau \neq 0$. Using the following identity,

$$\cos(A) \cos(B) = \frac{1}{2} \{\cos(A - B) + \cos(A + B)\}$$

we have

$$\mathbb{E}[\cos(\nu t + \Theta) \cos\{\nu(t + \tau) + \Theta\}] = \frac{1}{2} [\cos(\nu\tau) + \mathbb{E}[\cos\{\nu(2t + \tau) + 2\Theta\}]]$$

and

$$\begin{aligned} \mathbb{E}[\cos\{\nu(2t + \tau) + 2\Theta\}] &= \int_{-\pi}^{\pi} \cos\{\nu(2t + \tau) + 2\theta\} f_{\Theta}(\theta) d\theta \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos\{\nu(2t + \tau) + 2\theta\} d\theta + \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos\{\nu(2t + \tau) + 2\theta\} \cos(\theta) d\theta \\ &= \frac{1}{4\pi} \sin\{\nu(2t + \tau) + 2\theta\} \Big|_{-\pi}^{\pi} + \frac{1}{4\pi} \int_{-\pi}^{\pi} \cos\{\nu(2t + \tau) + \theta\} d\theta \\ &\quad + \frac{1}{4\pi} \int_{-\pi}^{\pi} \cos\{\nu(2t + \tau) + 3\theta\} d\theta \\ &= 0 + \frac{1}{4\pi} \sin\{\nu(2t + \tau) + \theta\} \Big|_{-\pi}^{\pi} + \frac{1}{12\pi} \sin\{\nu(2t + \tau) + 3\theta\} \Big|_{-\pi}^{\pi} \\ &= 0 + 0 + 0 \\ &= 0. \end{aligned}$$

We finally get $\mathbb{E}[X_t X_{t+\tau}] = \mathbb{E}[\epsilon_t \epsilon_{t+\tau}] \cos(\nu\tau) / 2 = \sigma^2 / 2$ if $\tau = 0$ and 0 otherwise, and therefore X_t is stationary.